Dataset and setup: SST-2

1 SST-2: binary sentiment

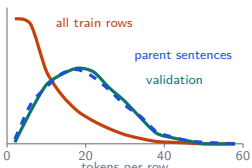
movie-review sentences, negative or positive

67,349 train rows

872 validation

1,821 test (hidden labels)

Sentence length



Every parse-tree node is a train row

neg comes across as a fairly weak retooling .
neg comes across as a fairly weak retooling
pos comes
neg as a fairly weak retooling
neg a fairly weak retooling
neg fairly weak retooling
neg fairly weak
neg weak

2 Configuration

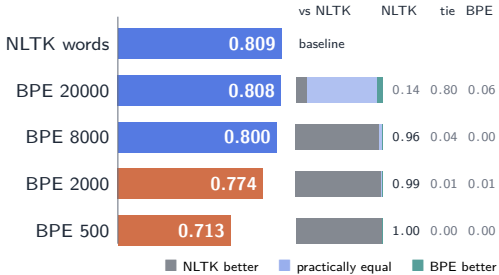
Choice	Setting
Tokenizers	NLTK words; BPE (trained on SST-2) with 500, 2k, 8k and 20k tokens
Features	token indicators, bag of words, mean GloVe
Models	log-linear, MLP, linear SVM, tuned MLP, 2-layer MLP, tuned C SVM
MLP	32 units, dropout 0.3, AdamW (lr 1e-4, wd 1e-2), 10 epochs
SVM	linear, $C = 1$
Evaluation	5-fold CV on whole-sentence folds, 3 seeds for NLTK systems (1 for BPE); accuracy
Analysis	Bayesian correlated t-test, ROPE of 0.005

Train is 6,885 sentences with their sub-phrases; 77% share the sentence's label · Validation is balanced, so accuracy is a valid metric (baseline 0.509) · 371 duplicate rows, 5 conflicting labels, no overlap with validation

Tokenization

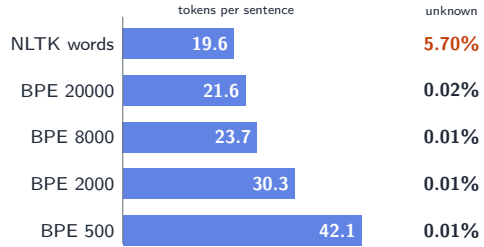
1 Accuracy by tokenizer

log-linear, token indicators, CV on parent folds, ROPE 0.005



2 Vocabulary and coverage

validation; AG News BPE gives 26% more chunks, so we train on SST-2



BPE 20000 is almost the same as NLTK · BPE 500, 2000 and 8000 are worse than NLTK · We use NLTK

Representations: indicators, counts and GloVe

Example sentence: ***not good , not bad***

1 Token indicators

does the token occur? (sparse; 14,802 dimensions)

,	bad	good	great	not
1	1	1	0	1

0.809 CV, log-linear, 3 seeds

2 Bag of words

how often does it occur? (sparse; 14,802 dimensions)

,	bad	good	great	not
1	1	1	0	2

0.808 CV, log-linear, 3 seeds

3 Mean GloVe (ours)

average of the word vectors (dense; 300 dimensions)

first 12 of 300 values

0.787 CV, log-linear, 3 seeds

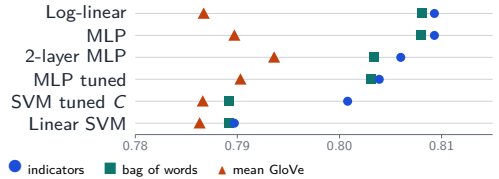
4 Bayesian comparison (log-linear, ROPE 0.005)

Indicators and bag of words are practically equivalent: only 7.3% of features have a count above 1 · Mean GloVe is worse: averaging loses word order · Considered but not tried: TF-IDF, n-grams, a learned embedding layer

Architectures: log-linear, MLP and linear SVM

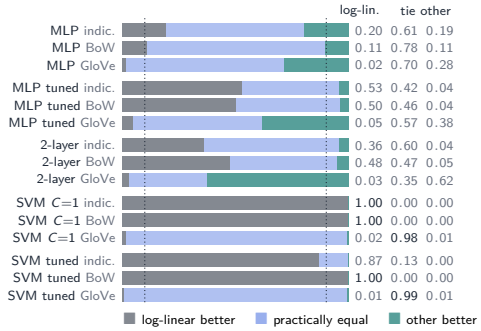
1 Mean CV accuracy

NLTK, 15 folds; MLP 32 units, dropout 0.3, lr 10^{-4} (tuned: 128 units, lr 10^{-3})

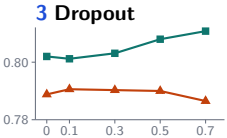


2 Log-linear vs every other model

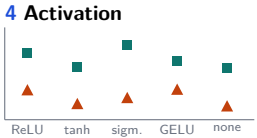
Bayesian correlated t-test, ROPE 0.005, same features



3 Dropout



4 Activation

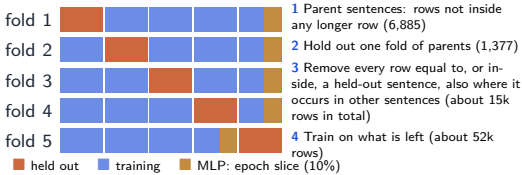


A hidden layer gives no evidence of improvement · Tuning the MLP or adding a second layer gives no reliable gain · Log-linear beats the C=1 SVM on indicators and bag of words; tuning C on indicators lifts it from 0.790 to 0.801 ($P = 0.94$)

Evaluation: cross-validation without leakage

1 Parent folds

how the training rows of one fold are built (5 folds, 3 seeds)



1 Parent sentences: rows not inside any longer row (6,885)

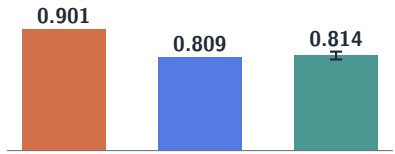
2 Hold out one fold of parents (1,377)

3 Remove every row equal to, or inside, a held-out sentence, also where it occurs in other sentences (about 15k rows in total)

4 Train on what is left (about 52k rows)

2 Random folds leak

86.5% of held-out rows lie inside a training row; log-linear, NLTK, indicators



error bar: standard error of 872 sentences (0.013)

3 Why a ROPE of 0.005

accuracy differences on a log scale; shaded: any value from 0.003 to 0.01 would be defensible

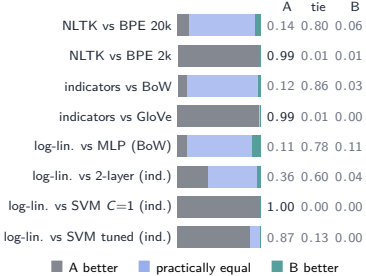
Random folds give higher CV accuracy through leakage · 3 seeds × 5 folds = 15 paired scores · Building the tokenizer on all of train causes no inflation (per-fold vocabulary changes at most 0.003)

Analysis and choice

Choice: log-linear with token indicators on NLTK tokens: CV 0.809, tied with the MLPs, simplest, fastest

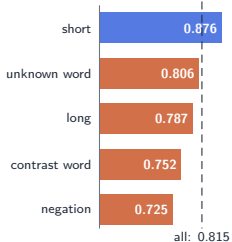
1 Pairwise results

ROPE 0.005; tokenizer rows: 1 seed



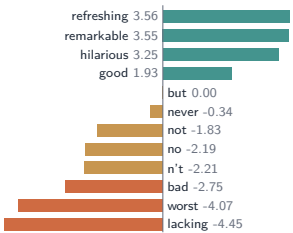
2 Hard sentences

accuracy of the MLP on validation



3 Each word gets one fixed weight

log-linear, bag of words



No clear winner among log-linear and MLPs on indicators and counts: undecided at ROPE 0.005, practically equivalent at 0.01 · 5 of 6 systems predict 53–55% positive (50.9% true) · MLPs are 3–7 times slower, for no gain